

NETWORK CITY

PORTS & PROTOCOLS

A colorful guide to service doors and
data delivery



LEARN THE SERVICE. FIND THE PORT. SOLVE THE SCENARIO.

Detailed comic illustrations + short definitions + clear exam clues

Start with the pictures. Learn each term in plain language. Then match each service to its default port and transport. This guide covers objective 2.1, with secure variants and useful real-world context.

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- 03 TCP: the careful delivery team
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Independent study guide. Aligned to exam 220-1201; not an official CompTIA publication. Illustrations are memory aids. Technical notes and labeled diagrams explain the real rules.

Ports: find the right door

VISUAL: Draw a neon-blue server city. Gold couriers choose glowing service doors beside a purple router officer.



Right building. Right door. Right service!

KNOCK!

- **Protocol.** A shared set of rules for sending and reading data. Think of a game: both sides must know the rules.

- **IP - Internet Protocol.** IP helps move packets, or labeled data parcels, between networks. An IP address identifies a network interface. In our story, it is the building address.

- **Port number.** A number used by TCP or UDP to help deliver data to the right program or conversation. It is a software door, not a cable socket.

- **Client / server.** A client asks for a service. A server provides it. One computer can run several services, each listening for traffic at its own endpoint.

- **Two door lists.** TCP port 53 and UDP port 53 are separate endpoints. Always name the transport as well as the number.

- **Number ranges.** Each transport has ports 0-65535. IANA, the Internet Assigned Numbers Authority, groups 0-1023 as system ports, 1024-49151 as user ports, and 49152-65535 as dynamic/private ports. Port 0 is reserved. [2]

- **Default / ephemeral.** Default means the usual setting; it can change. Ephemeral means temporary: clients often use temporary source ports. The exact client range depends on the operating system.

EXAM TARGET: Core 1, 220-1201, objective 2.1. Learn each service's purpose, default port, and TCP/UDP behavior. A port number alone does not prove which program is using it. [1, 2]

TCP: the careful delivery team

VISUAL: Draw a cyan delivery robot with a checklist and receipt. Its truck carries ordered parcels along orange roads. 



I track the bytes and put them in order!

CHECK!

- **TCP - Transmission Control Protocol.** A transport protocol that gives applications a reliable, ordered stream of bytes. A byte is a small unit of data, usually eight bits. [3]
- **Connection-oriented.** The two endpoints set up a connection before normal data exchange. Like two callers saying hello before they talk.
- **Three-way handshake.** Client: SYN. Server: SYN-ACK. Client: ACK. SYN means synchronize; ACK means acknowledgment. They agree on starting sequence numbers.

- **Sequence number.** A position marker for bytes. TCP can put received data in the correct order even when pieces take different paths.
- **Acknowledgment / retransmission.** The receiver reports progress. The sender can resend missing data. Retransmission means sending again.
- **Flow / congestion control.** Flow control helps avoid overwhelming the receiver. Congestion control adjusts sending when the network path is crowded.
- **Use cases.** File transfers, email, and remote commands benefit from ordered delivery. TCP can still time out when a connection fails; it cannot fix a broken cable by magic.

RELIABLE DOES NOT MEAN SECRET. TCP does not encrypt data by itself. Reliability costs extra tracking and may add delay when data must be resent.

UDP: send the message now

VISUAL: Draw a purple rocket courier racing toward a live video screen. One late parcel falls behind the glowing road.



Each message gets its own quick launch!

WHOOSH!

Feature	TCP	UDP
Data style	Ordered byte stream	Separate datagrams
Built-in recovery	Acknowledgments; retries	No delivery retries
Connection setup	Handshake	No TCP-style handshake
Common examples	SSH; FTP; SMTP	DHCP; many DNS queries

- UDP - User Datagram Protocol.** A transport protocol that sends separate messages called datagrams. UDP itself does not set up a TCP-style connection. [4]
- Connectionless.** Each datagram can be sent without a transport handshake. There is no built-in promise of delivery, order, or protection from duplicate messages.
- Why use it?** Less built-in tracking can help short requests and live traffic. An old voice sample may be less useful than the next fresh sample.
- Application rules still matter.** Programs can add retries, ordering, or other safeguards above UDP. A DNS client, for example, can retry an unanswered query.

SPEED TRAP: UDP is not always faster in real use. Neither TCP nor UDP has one fixed Mbps (megabits per second) rating or cable-distance limit. Hardware, links, traffic, and application design set the result.

FTP: commands and cargo

VISUAL: Draw a green file warehouse with a small control window and a separate wide cargo door. Gold folders ride conveyors. Gold folders ride conveyors.



One connection talks. Another moves files!

LIFT!

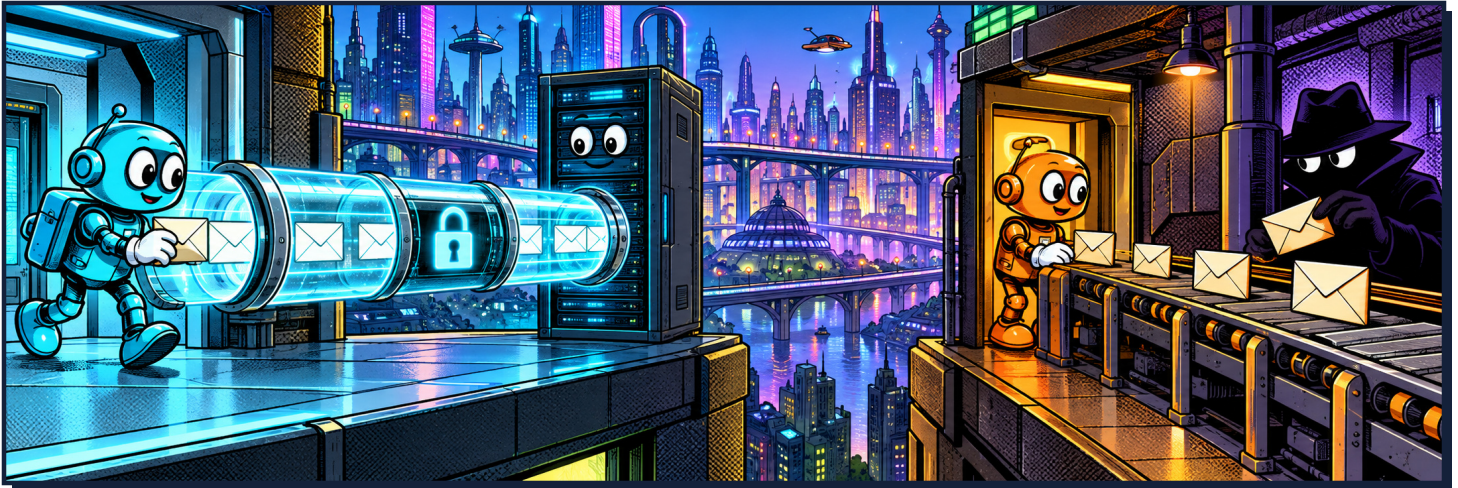
- **FTP - File Transfer Protocol.** A protocol for uploading and downloading files. Upload sends a file to another system. Download receives a file. [5]
- **TCP 21 - control.** The client opens the command connection to server port 21. Commands handle tasks such as logging in or selecting a file.
- **TCP 20 - active data.** In classic active FTP, the server normally starts the data connection from its port 20 to a client-specified port. File contents and directory listings use a separate data connection.

- **Passive FTP.** The server tells the client which server data port to contact. The client starts that connection. Passive data ports are negotiated; they are not always port 20.
- **Why two paths?** Instructions and file data have different jobs. The clerk takes your order while the loading door handles the boxes.
- **Use case.** Transfer files to a system that requires FTP. Firewalls must account for both control and data behavior; opening only 21 may not make transfers work.
- **Plaintext.** Ordinary FTP does not encrypt its commands, login details, or files. Plaintext means the content is readable without decrypting it.

MEMORY HOOK: 21 = the talking clerk; 20 = classic active-mode cargo. SFTP is a different protocol over SSH, usually TCP 22. It is not ordinary FTP with a new name.

SSH & Telnet: remote commands

VISUAL: Draw a cyan command courier inside a locked tunnel. Beside it, an orange courier sends readable notes on an open belt.



Same kind of job. Different protection!

TYPE!

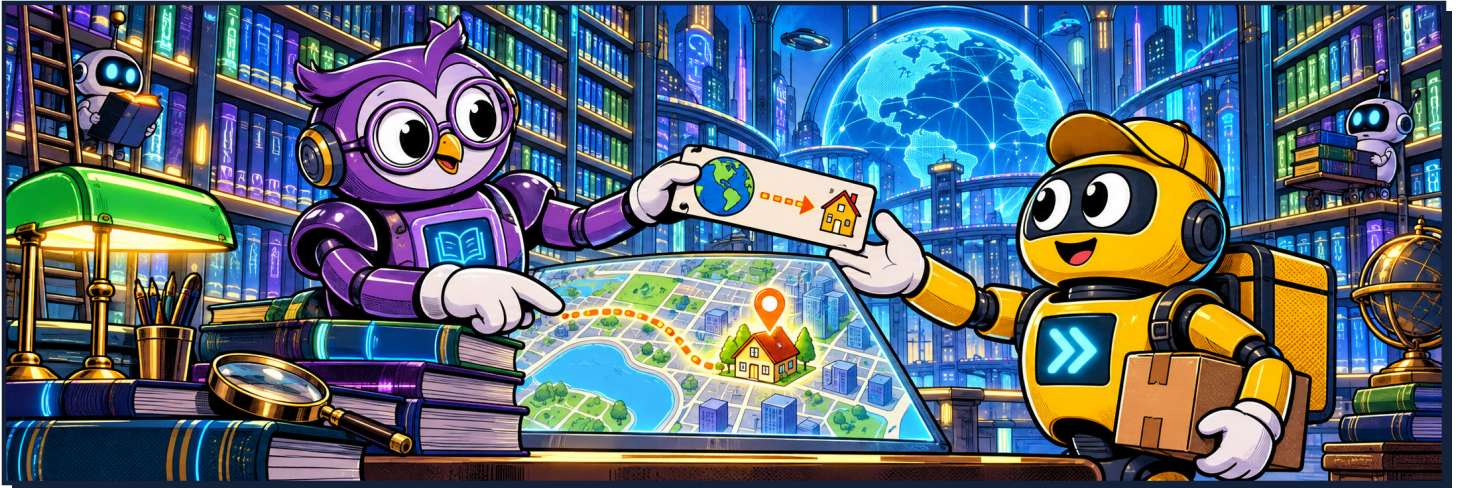
- **Remote access.** Using one computer to work on another through a network. A command-line interface lets you type instructions instead of clicking desktop icons.
- **SSH - Secure Shell.** Usually TCP 22. SSH protects remote login and command sessions with encryption. It also supports tasks such as secure file transfer. [6]
- **Encryption.** Scrambles data so an observer cannot read it without the right key. Authentication checks who or what is connecting.

- **Host key.** A server identity key used by SSH. Verify unexpected key changes; the key helps the client recognize the intended server.
- **Telnet.** A legacy remote terminal protocol, usually TCP 23. Telnet is the protocol's name; learn its remote-terminal job. Basic Telnet sends session data without encryption. [7]
- **Use cases.** Choose SSH for secure text-based administration when supported. Telnet may appear on old equipment or in a controlled training lab.
- **Exam clue.** "Encrypted remote shell" points to SSH 22. "Unencrypted remote terminal" points to Telnet 23. Neither term means a full graphical desktop.

22: SSH protects the shell. 23: Telnet exposes the text. Do not assume that moving Telnet to a different port adds encryption.

DNS: the name detective

●●● **VISUAL:** Draw a purple owl librarian matching a globe-name card to a location on a neon city map. A gold courier waits.



Give me the name. I will look up the record!

LOOK!

- **DNS - Domain Name System.** A distributed lookup system. It can turn a name such as `www.example.com` into an IP address and store other kinds of records.

- **Domain name / resolver.** A domain name is a human-friendly name. A resolver is the software or service that looks up DNS answers for a client.

- **Port 53 - UDP and TCP.** Many ordinary queries use UDP 53. DNS also uses TCP 53, including when a response is too large for the UDP exchange. DNS implementations must support TCP. [8]

- **Records.** An A record holds an Internet Protocol version 4 (IPv4) address. An AAAA record holds a version 6 (IPv6) address. An MX, or mail exchanger, record points toward mail servers. [21]

- **Cache / TTL.** A cache is a saved answer. TTL means Time to Live: it tells a cache how long it may keep that DNS record before it needs a fresh answer.

- **Why it helps.** People remember names more easily than long numbers. The address behind a name can change while the name stays familiar.

- **Trouble clue.** If a service works by address but fails by name, investigate DNS. That is a clue, not proof: some web services also need the correct host name.

DNS FINDS; DHCP ASSIGNS. DNS looks up records. DHCP supplies network settings. Remember 53 for DNS and both TCP and UDP.

DHCP: rent network settings

●●● **VISUAL:** Draw a lime-green hotel server lending address keycards to new laptops. A bright clock marks each temporary lease.



Here are your settings. Keep track of the lease!

DORA!

- **DHCP - Dynamic Host Configuration Protocol.**

Automatically supplies settings to network clients. For IPv4 DHCP, the server uses UDP 67 and the client uses UDP 68. [9]

- **Lease.** Permission to use an address for a limited time. Clients try to renew before it expires. A reservation can give one client a predictable address.

- **D - Discover.** A new client broadcasts to find a DHCP server. Broadcast means the message reaches the local broadcast area, rather than one known recipient.

- **O - Offer.** A server offers an address and settings. The client does not simply keep every offered address.

- **R - Request.** The client asks to use an offered address and signals which offer it chose.

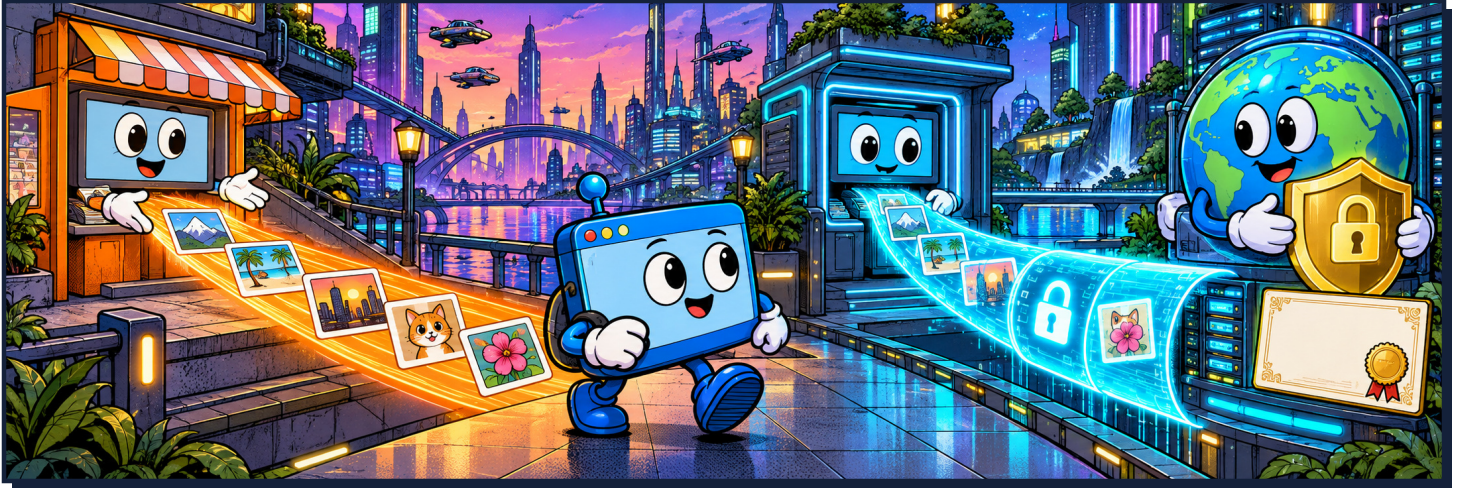
- **A - Acknowledgment.** The selected server confirms the lease. DORA is a memory aid for this common initial four-message exchange; renewals can use a shorter exchange.

- **Useful settings.** The subnet mask helps identify the local IPv4 network. The default gateway leads to other networks. DNS server addresses tell the client where to ask name questions.

67 = SERVER; 68 = CLIENT. This pair is for IPv4 DHCP. DHCPv6 uses different ports, 547/546; that is extra context, not the 67/68 pair in this objective. [23]

HTTP & HTTPS: web delivery

VISUAL: Draw a blue browser visiting an open orange web path and a cyan locked web tunnel. Gold picture cards carry page content.



Fetch the page. Check the connection!

CLICK!

- **HTTP - Hypertext Transfer Protocol.** Rules for web requests and responses. Hypertext means text that can link to other content. HTTP also carries images and application data. [10]
- **TCP 80 - HTTP.** The usual default for ordinary HTTP. A request might ask for a page; the server returns a response. HTTP alone does not encrypt the exchange.
- **HTTPS - HTTP Secure.** HTTP protected with TLS, or Transport Layer Security. The usual Core 1 pairing is HTTPS with TCP 443.

- **TLS.** Adds encryption, integrity checks, and server authentication when properly validated. Integrity means helping detect unwanted changes to data.
- **Certificate.** A digital identity document used in the TLS process. The browser checks matters such as the requested host name and whether the certificate is trusted.
- **What the lock means.** It helps protect the connection. It does not promise that the site owner is honest, the item for sale is real, or every download is harmless.
- **Version note.** HTTP/1.1 and HTTP/2 commonly use TCP. HTTP/3 uses QUIC over UDP, commonly UDP 443. QUIC includes secure transport features. [11]

CORE RECALL: HTTP 80; HTTPS 443. Neither number names a speed or distance. Port 443 can carry TCP or UDP traffic, depending on the protocol version.

SMTP: send and relay email

●●● **VISUAL:** Draw an orange mail courier moving envelopes from a laptop to purple mail servers across glowing city roads.



Outgoing mail? I carry it onward!

SEND!

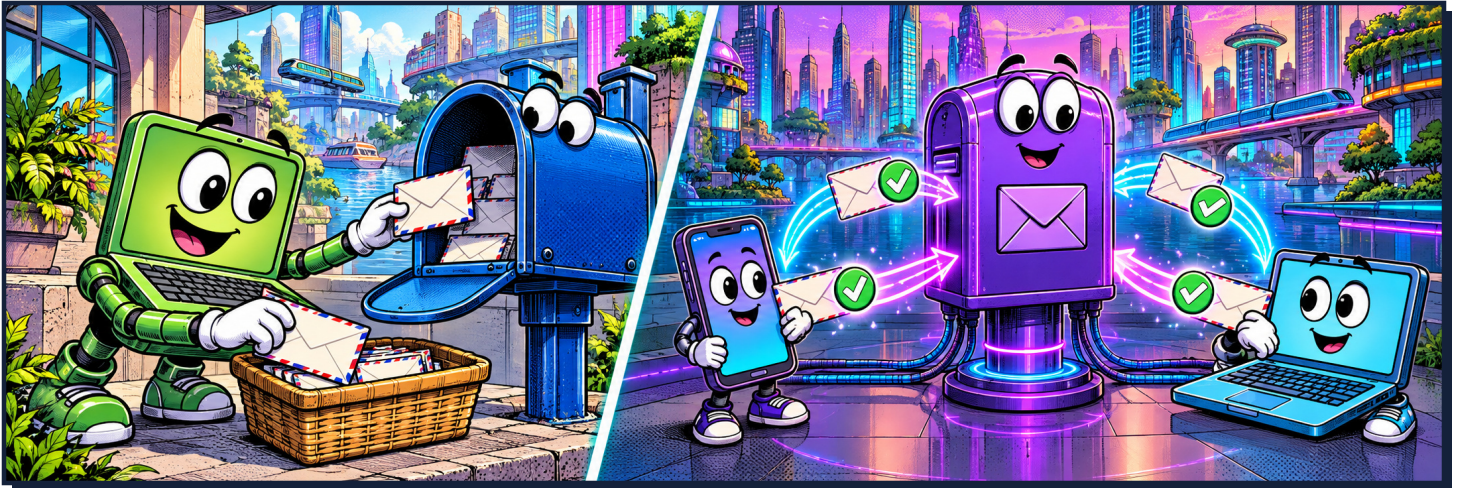
- **SMTP - Simple Mail Transfer Protocol.** The protocol for sending and relaying email. Relay means pass the message to another mail server. [12]
- **TCP 25.** The classic SMTP port, especially for mail-server-to-mail-server delivery. This is the SMTP port listed in the Core 1 objective.
- **A mail journey.** Your mail app submits a message to a mail service. That service can relay it to the recipient's mail service. A different protocol may then help the recipient read it.

- **Client submission.** A modern mail app commonly submits through TCP 587 with STARTTLS, or TCP 465 with implicit TLS. These are useful extra ports; do not replace exam SMTP 25 with them. [15]
- **STARTTLS.** A command that upgrades a supported connection to TLS. "Implicit TLS" starts TLS as soon as the connection begins, before normal mail commands.
- **Security nuance.** Port 25 traffic can use STARTTLS. The port number alone does not tell you whether a given SMTP conversation is encrypted.
- **Use case / clue.** If sending fails but reading works, check the sending service, login, and configured SMTP connection. POP3 or IMAP settings do a different job.

SMTP SENDS. POP3 AND IMAP HELP READ. Keep the direction straight: sending mail does not mean downloading the inbox.

POP3 & IMAP: open the inbox

●●● **VISUAL:** Draw a green laptop collecting mail into a basket. Beside it, a purple mailbox keeps a phone and laptop in sync.



Download a copy, or keep the mailbox in sync!

READ!

- **POP3 - Post Office Protocol version 3.** Usually TCP 110. A client retrieves messages from a server, often for local storage. It may leave copies on the server, depending on settings. [13]

- **POP3 use case.** A simple download-focused mail setup. Do not memorize “POP3 always deletes mail.” Deletion depends on the client’s actions and configuration.

- **IMAP - Internet Message Access Protocol.** Usually TCP 143. A client works with mailboxes kept on the server. It can synchronize folders, messages, and flags such as read/unread. [14]

- **IMAP use case.** Read the same mailbox from a phone and laptop. Server-side state helps both devices show matching folders and read status.

- **TLS choices.** POP3 over implicit TLS commonly uses 995. IMAP over implicit TLS commonly uses 993. Ports 110 and 143 can also support upgrades to TLS; they are not proof of encryption. [15]

- **Naming note.** The standard expands IMAP as Internet Message Access Protocol. Some study lists use “Internet Mail Access Protocol”; the port and mailbox job still identify IMAP.

MEMORY HOOK: POP3 110 = pick up mail. IMAP 143 = manage the shared inbox. SMTP 25 = move outgoing mail between servers.

NetBIOS: the legacy trio

VISUAL: Draw three retro neon service booths: a name directory, a message thrower, and a connected telephone pair.



Three neighboring doors. Three different jobs!

RETRO!

- **NetBIOS.** Network Basic Input/Output System. A legacy set of networking services used by applications. It is not the BIOS firmware that starts a PC.

- **NetBT.** NetBIOS over TCP/IP. This carries NetBIOS services using TCP/IP networking. TCP/IP names the wider Internet protocol family, not just one application.

- **137 - name service.** Registers or looks up NetBIOS names. UDP 137 is common; the name service also defines TCP 137. [16]

- **138 - datagram service.** Uses UDP. Sends connectionless messages. Think of the middle booth tossing separate postcards.

- **139 - session service.** Uses TCP. Provides a connection for a continuing conversation. Older Windows file sharing can carry SMB over this service.

- **Use case.** Recognize older Windows-network dependencies and troubleshoot legacy equipment. Modern direct-hosted SMB uses TCP 445 and does not need the NetBIOS trio. [18]

- **Name versus address.** A NetBIOS name is a legacy device/service name. Do not treat NetBIOS name lookup as a replacement term for every DNS query.

137 = NAMES. 138 = DATAGRAMS. 139 = SESSIONS. Learn the jobs with the numbers; do not label all three as “TCP only” or all three as “UDP only.”

LDAP: the directory desk

VISUAL: Draw a purple librarian server opening a glowing address book of people, groups, and computers. A cyan robot requests a record.



Find the right entry in the directory!

FIND!

- **LDAP - Lightweight Directory Access Protocol.** A protocol for reading and managing directory entries. A directory stores organized information about things such as users, groups, and computers. [17]

- **TCP 389.** The usual connection for LDAP requests and responses. Clients can search for entries or read selected attributes.

- **Entry / attribute.** An entry is one record, like a person's card. An attribute is one fact on it, such as a display name or email address.

- **AD - Active Directory.** Microsoft's directory system. LDAP is one of the protocols used to work with it. LDAP is a protocol; Active Directory is a system that provides directory services.

- **UDP 389 nuance.** Connectionless LDAP can use UDP 389 for specific tasks, including Active Directory domain-controller discovery. Do not assume all LDAP uses UDP or that UDP 389 never exists. [19]

- **Protection.** An LDAP connection on 389 can upgrade with StartTLS. LDAPS, LDAP over TLS, commonly uses TCP 636. Check configuration and certificate validation.

- **Use case.** A company app looks up employee information or group membership. Authentication checks identity; authorization decides what that identity may do.

LDAP = DIRECTORY ENTRIES. SMB = SHARED FILES. Both can involve a company server, but they solve different problems.

SMB: the shared file counter

VISUAL: Draw a lime-green file server sharing folders and a printer with cheerful laptops across cyan paths in a purple office.



One shared folder. Several allowed teammates!

SHARE!

- **SMB - Server Message Block.** A protocol used for shared files, printers, and related network services. A server can make a folder available to authorized clients. [18]

- **TCP 445.** The usual port for direct-hosted SMB. Direct-hosted means SMB runs directly over TCP without the older NetBIOS session service.

- **CIFS - Common Internet File System.** A name associated with an older SMB dialect. A dialect is a protocol version or variation. CIFS is not the name for every modern SMB version.

- **Share / permissions.** A share is a resource made available over the network. Permissions define who can read, create, or change its contents.

- **UNC - Universal Naming Convention.** A Windows network-path format. For example, \\fileserver\classwork names the classwork share on fileserver.

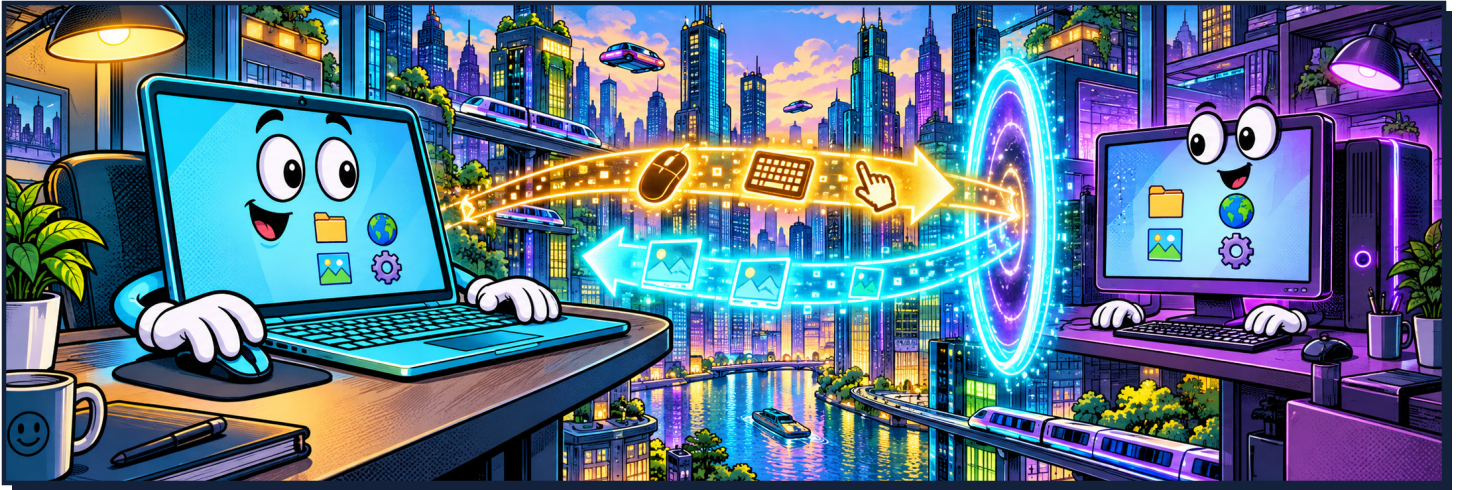
- **Use case.** Students open files from a classroom server. Workers save documents to an approved team share or use a shared printer.

- **Security / legacy clue.** Modern SMB supports security features such as encryption, depending on version and settings. Port 445 alone does not guarantee encryption. Older SMB may use NetBIOS TCP 139.

445 = DIRECT SMB FILE SHARING. Keep SMB versions current and grant only needed access. A blocked port, unavailable share, or missing permission can each stop a file request.

RDP: your faraway desktop

VISUAL: Draw a cyan laptop pilot controlling a distant purple desktop through a glowing screen portal. Gold keyboard and mouse signals travel across.



See the desktop. Work from another device!

REMOTE!

- **RDP - Remote Desktop Protocol.** A protocol for remote graphical sessions. You see a desktop and send keyboard and mouse actions over the network.
- **3389 - TCP and UDP.** RDP commonly uses TCP 3389 and can also use UDP 3389 for transport performance. Exact behavior depends on the client, server, and configuration. [20]
- **Client / host.** The client starts the remote session. The host provides the desktop. The host must support and enable the service; not every operating-system edition can host it.
- **Use case.** An authorized technician manages a distant Windows computer, or a worker uses a desktop provided by their organization.
- **RDP versus SSH.** RDP gives a graphical desktop. SSH is best known for a secure text shell. Both support remote work, but the exam clues differ.
- **Safe access.** Use approved accounts and protected access methods. Organizations may require a VPN, or Virtual Private Network, or a Remote Desktop Gateway. Avoid casually exposing a desktop service to the public internet.
- **Trouble clue.** Check that the host is awake, the service is enabled, access is allowed, and the network path permits the configured connection.

3389 = REMOTE DESKTOP. Changing the port does not replace authentication, updates, or proper access rules. Always get authorization before controlling another computer.

Port map A: web, files & setup

●●● **VISUAL:** Draw gold packet couriers lining up at neon-blue service doors. A purple router holds a colorful city map.



Say the job, the number, and the transport!

RECALL!

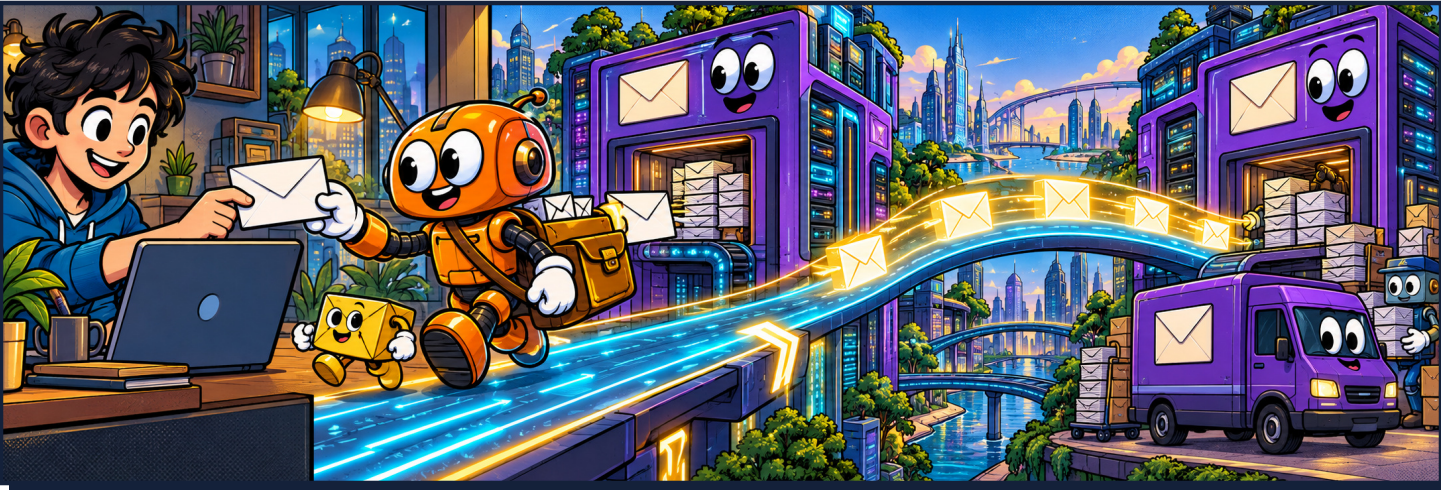
Service	Default port	Transport / job
FTP	20 / 21	TCP: active data / control
SSH	22	TCP: secure remote shell
Telnet	23	TCP: legacy remote terminal
DNS	53	UDP and TCP: name records
DHCP (IPv4)	67 / 68	UDP: server / client
HTTP	80	TCP: web traffic without TLS
HTTPS	443	TCP commonly; HTTP/3 uses UDP

These are normal service defaults, not permanent laws. Administrators can change listening ports. FTP passive-mode data uses a negotiated server port. [1-11]

PRACTICE: Cover the middle column. Can you name every port from the service alone? Then cover the first column and work backward.

Port map B: mail & company tools

VISUAL: Draw orange mail couriers passing purple offices while friendly server clerks prepare inboxes, files, and directory records.



Match the service to the problem!

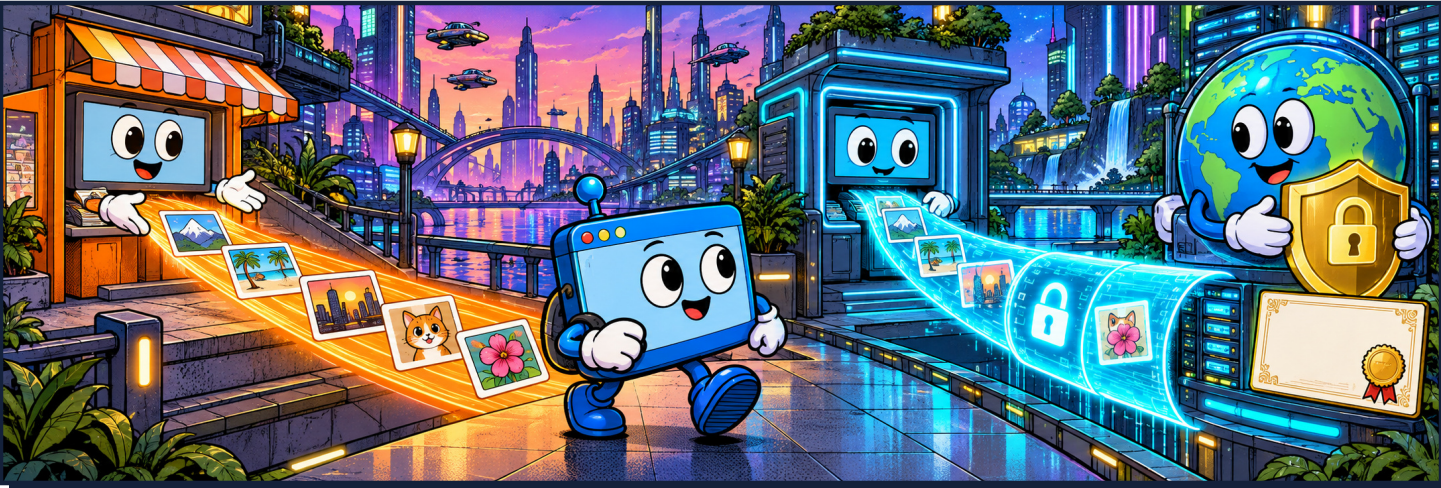
MATCH!

Service	Default port	Transport / job
SMTP	25	TCP: send / relay email
POP3	110	TCP: retrieve email
IMAP	143	TCP: manage server mailbox
NetBIOS names	137	Usually UDP; TCP also defined
NetBIOS datagrams	138	UDP: connectionless messages
NetBIOS sessions	139	TCP: legacy sessions
LDAP	389	Usually TCP; UDP for specific uses
SMB / CIFS	445	TCP: direct file / printer sharing
RDP	3389	TCP and UDP: remote desktop

OBJECTIVE 2.1 CHECK: The two port maps cover every service in the official list. Secure variants on the next page add practical context. [1, 12-20]

Secure extras & exam traps

●●● **VISUAL:** Draw cyan locks protecting web and mail pathways. A gold detective checks each service instead of trusting its door number.



A familiar port is a clue, not proof!

VERIFY!

Extra service	Usual TCP port	Meaning
Mail submission	587	SMTP; commonly upgraded with STARTTLS
Mail submission with TLS	465	SMTP with implicit TLS
IMAP with TLS	993	Encrypted mailbox access
POP3 with TLS	995	Encrypted mail retrieval
LDAP with TLS	636	Encrypted directory access

- SFTP versus FTPS.** SFTP means SSH File Transfer Protocol, usually TCP 22. FTPS means FTP protected with TLS; explicit FTPS commonly starts on TCP 21. These are different protocols. [22]
- No magic port.** A service may use a nondefault port. Firewall rules need the correct transport and direction as well as the number.

EXTRA CONTEXT: These secure variants are not additional named ports in objective 2.1. Learn the main list first. “Uses TCP” does not mean “encrypted.” [1, 6, 15, 17]

Mission: load a secure website

VISUAL: Draw a new gold courier collecting a keycard, asking the purple directory owl, then entering a cyan web door.



Get settings. Find the address. Request the page!

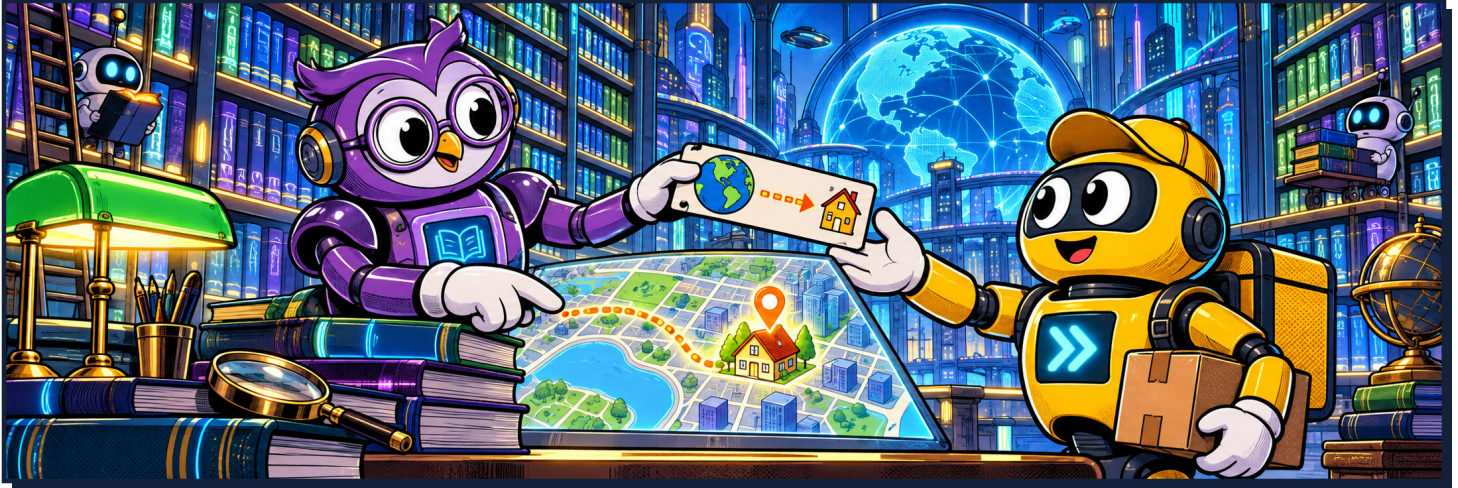
GO!

- **1. Get settings.** A newly connected client may use DHCP: UDP 68 on the client, UDP 67 on the server. It learns an address, gateway, and DNS server settings.
- **2. Find the destination.** The browser needs the website address. If no usable cached answer exists, a DNS lookup can help find it. Ordinary DNS uses port 53.
- **3. Connect securely.** For a typical HTTPS-over-TCP session, the browser opens a TCP connection to server port 443, negotiates TLS, then sends HTTP requests inside that protection.
- **4. Keep replies sorted.** The client also has a source port. Example: 192.168.1.25:51514 → 203.0.113.10:443. The numbers after the colons are ports; these addresses illustrate a flow.
- **5. Read the clues.** The same server IP can offer several services. Source/destination IP addresses, source/destination ports, and transport help distinguish conversations.
- **Troubleshoot in order.** Check link and IP settings, name lookup, the actual service, then relevant firewall rules and application settings. A successful ping does not prove that a web or mail service works.
- **No physical range.** Software ports are not jacks. Cable category, radio standard, and equipment determine physical reach and link rates; port 443 does not mean 443 Mbps.

SCENARIO RULE: Choose the service by its job first. Then match its default port and transport. Do not turn off every security control just to make a connection work.

Sources & exam alignment

● ● ● **VISUAL:** Draw the purple directory owl beside neon shelves of reference books. Gold couriers return their checked fact cards.



The story is playful. The facts have sources!

CHECK!

Primary scope: CompTIA A+ Core 1, 220-1201, objective 2.1. References are primary technical sources. RFC means Request for Comments, a technical document series.

1. **CompTIA: Core 1 objectives, 2.1**
2. **IANA: Service names and ports**
3. **RFC 9293: TCP**
4. **RFC 768: UDP**
5. **RFC 959: FTP**
6. **RFC 4251: SSH architecture**
7. **RFC 854: Telnet**
8. **RFC 7766: DNS over TCP**
9. **RFC 2131: DHCP**
10. **RFC 9110: HTTP semantics**
11. **RFC 9114: HTTP/3**

12. **RFC 5321: SMTP**
13. **RFC 1939: POP3**
14. **RFC 9051: IMAP**
15. **RFC 8314: Mail TLS**
16. **RFC 1002: NetBIOS transport**
17. **RFC 4511: LDAP**
18. **Microsoft: SMB and CIFS**
19. **Microsoft: AD connectivity / LDAP UDP**
20. **Microsoft: Remote Desktop ports**
21. **RFC 1035: DNS records and caching**
22. **RFC 4217: FTP with TLS**
23. **RFC 8415: DHCPv6**

Reviewed September 17, 2026. Independent study guide; not endorsed by CompTIA. Artwork uses teaching metaphors and was made with the built-in image tool.

STAR QUIZ TIME

●●● **VISUAL:** Draw the neon service city ready for a three-star challenge. Gold couriers cheer beside the purple router officer.



You have the map. Now solve the mission!

STAR!

- 1. Find the name!** A computer needs a website's IP address. Which service should it ask, which port does it use, and is it UDP-only?
- 2. Get a lease!** A new IPv4 client needs automatic network settings. Name the protocol and the server/client UDP ports. What does DORA stand for?
- 3. Keep the inbox in sync!** A learner wants matching folders and read status on a phone and laptop. Pick POP3, IMAP, or SMTP. Give its base port and common implicit-TLS port.

ANSWERS - CHECK YOUR HERO SKILLS!

- 1. DNS, port 53, both UDP and TCP.** Great detective work! DNS looks up name records. Many queries use UDP, but TCP is part of normal DNS support too.
- 2. DHCP; server UDP 67, client UDP 68.** DORA means Discover, Offer, Request, Acknowledgment. Nice work! The server lends settings through a lease.
- 3. IMAP; TCP 143 and TCP 993.** You nailed the mailbox job! IMAP manages server-side mailboxes and shared state. SMTP sends mail; POP3 focuses on retrieval.